

PSY310- Lab in Psychology

Section 1

Lab Report 3-Visual search

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Git hub link:

<https://github.com/aviks-afk/PSY-310-LAB-IN-PSYCHOLOGY/tree/visual-search>

Lab Report: Visual Search Task

Introduction

The visual search experiment was initially characterized by Treisman and Gelade (1980) and is one of the basic paradigms in cognitive psychology that studies how people can find a target stimulus in a set of distractors. The task clarifies the mechanisms of attention, perception and decision making that are undertaken in filtering the appropriate visual information out of distracting information. The purpose of this study is to examine the influence of the number of distractors on response time in visual search task with the conditions of 5 and 10 distractors.

Research Questions

Are the distractors (5 vs. 10) selected on the type of response time that participants make when they are searching a target?

Research Objectives

- To compare the average response times between two series of trials that were conducted with 5 distractors and two series of trials that were conducted with 10 distractors.
- To determine the effect of the increasing distractors on the speed of cognitive processing in a visual search.

Hypothesis

The participants will show shorter response times in the 5 distractors than the 10 distractors.

Method

Participants

The data is 100 answers of the participant who performs the visual search trials with different numbers of distractors.

Materials

- PsychoPy software.
- Reactions were provided by clicking on the mouse that was close to the target.
- Response times were taken to be the time between the beginning and the end of each trial.

Procedure

The participants were educated about the task and were provided with the privilege of dropping out at any time. A fixation cross was used to initiate each of the trials, then the target and distractors appeared. The subjects reacted by clicking the mouse key in the vicinity of the target. Trials were of either 5 or 10 distractors. The response time and accuracy was recorded.

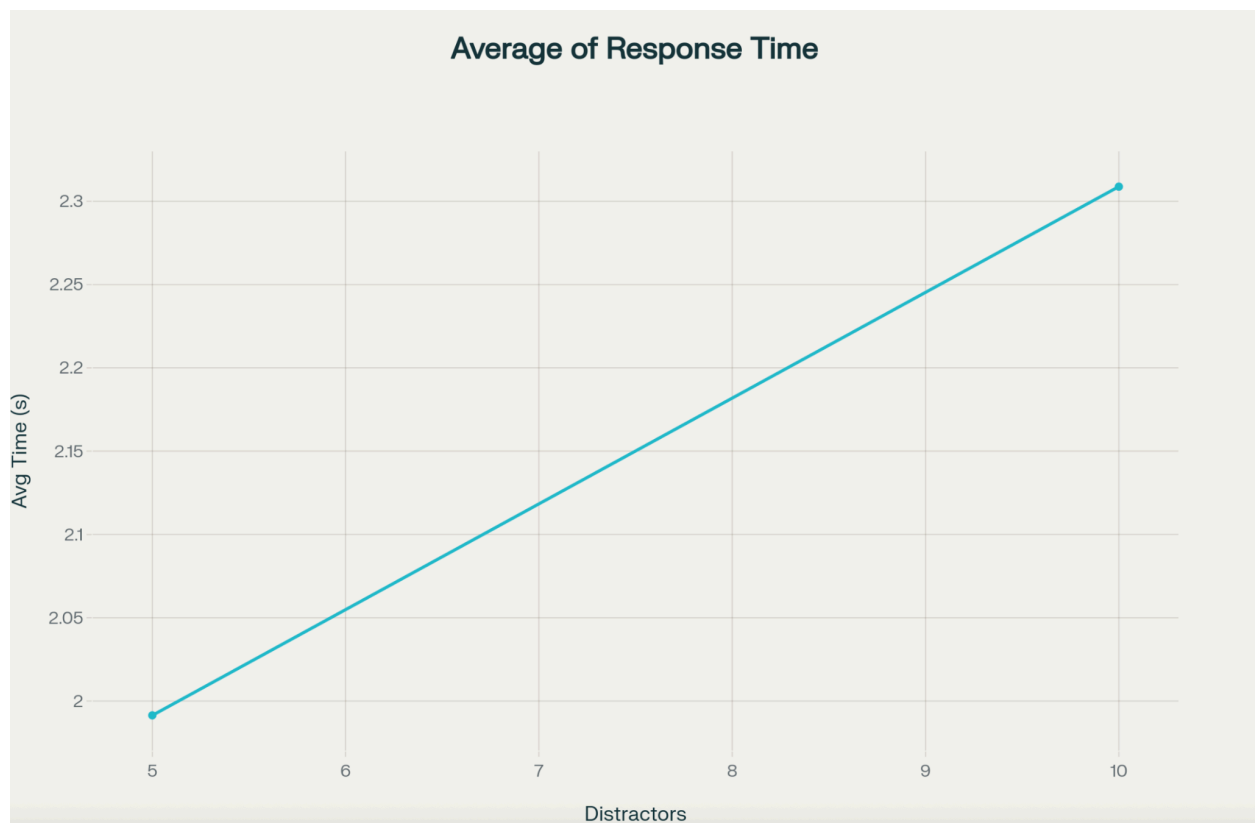
Results

Table 1: Calculation of Average Response Time

No. of Distractors	Average Response Time (seconds)	Standard Deviation	Number of Trials
5	1.9914	0.5974	49
10	2.3088	0.9226	51

Graph 1: Average of Response Time

Average Response Time Line Graph:



Graph 2: Slope (Distractor $\times$ Average Response Time)

Slope Line Graph



Discussion

This experiment proves that the more the distractors, the more the response time in the process of visual search grows. The results are consistent with the current attention theory by Treisman and Gelade (1980) in which more complex distractor complexity slows down search efficacy. The standard deviation of the trials having 10 distractors is greater and this indicates that there is more variation in difficulty. Increase of the cognitive load under more demanding conditions is also revealed by the slope analysis.

Limitations

- External validity is inhibited by individual participant data.
- The complexities of search in the real-life may not be recreated in controlled conditions.
- Future research ought to increase sample size and have more diverse stimuli.

References

Treisman, A. M., Gelade, G. (1980). A feature-integration theory of attention. *Cognitive Psychology*, 12(1), 97-136.

Peirce, J., Gray, J. R., Simpson, S., MacAskill, M., et al. (2019). PsychoPy2: Experiments in behavior made easy. *Behavior Research Methods*, 51(1), 195-203.